# Assignment 5 soln

**Part #1**

```
module top();
    bit clk;

    initial begin
        forever
            #1 clk = ~clk;
    end

    ALSU_if ALSUif (clk);
    ALSU DUT (ALSUif.A, ALSUif.B, ALSUif.cin, ALS

    initial begin
        run_test("ALSU_test");
    end

endmodule
```

```
class ALSU_env extends uvm_env;
  `uvm_component_utils(ALSU_env)

  function new(string name = "ALSU_env", uvm_component parent = null);
    super.new(name,parent);
  endfunction
endclass
```

KareemWaseem

```
class ALSU_test extends uvm_test;
  `uvm_component_utils(ALSU_test)

  ALSU_env env;

  function new(string name = "ALSU_test", uvm_component parent = null);
    super.new(name, parent);
  endfunction

  function void build_phase(uvm_phase phase);
    super.build_phase(phase);
    env = ALSU_env::type_id::create("env",this);
  endfunction

  task run_phase(uvm_phase phase);
    super.run_phase(phase);
    phase.raise_objection(this);
    #100;
    `uvm_info("run_phase", "Welcome to the UVM Env.", UVM_MEDIUM)
    phase.drop_objection(this);
  endtask
endclass: ALSU_test
```

**Part #2**

```
module top();
    bit clk;

    initial begin
        forever
            #1 clk = ~clk;
    end

    ALSU_if ALSUif (clk);
    ALSU DUT (ALSUif.A, ALSUif.B, ALSUif.cin, ALSUif.serial_in, ALSUif.red_op_A, ALSUif.r

    initial begin
        uvm_config_db#(virtual ALSU_if)::set(null, "uvm_test_top", "ALSU_IF", ALSUif);
        run_test("ALSU_test");
    end

endmodule
```

Kareem Waseem

```
class ALSU_test extends uvm_test;
  `uvm_component_utils(ALSU_test)

  ALSU_env env;
  virtual ALSU_if ALSU_if_vif;

  function new(string name = "ALSU_test", uvm_component parent = null);
    super.new(name, parent);
  endfunction

  function void build_phase(uvm_phase phase);
    super.build_phase(phase);
    env = ALSU_env::type_id::create("env",this);

    if (!uvm_config_db #(virtual ALSU_if)::get(this, "","ALSU_IF", ALSU_if_vif))
      `uvm_fatal("build_phase", "Test - Unable to get the virtual interface of the ALSU from the uvm_config_db");

    uvm_config_db #(virtual ALSU_if)::set(this, "*", "ALSU_IF_comps", ALSU_if_vif);
  endfunction

  task run_phase(uvm_phase phase);
    super.run_phase(phase);
    phase.raise_objection(this);
    #100;
    `uvm_info("run_phase", "Welcome to the UVM Env.", UVM_MEDIUM)
    phase.drop_objection(this);
  endtask
endclass: ALSU_test
```

```
class ALSU_env extends uvm_env;
    `uvm_component_utils(ALSU_env)

  ALSU_driver driver;

  function new(string name = "ALSU_env", uvm_component parent = null);
    super.new(name,parent);
  endfunction

  function void build_phase(uvm_phase phase);
    super.build_phase(phase);
    driver = ALSU_driver::type_id::create("driver",this);
  endfunction: build_phase
endclass
```

Kareem Waseem

```
class ALSU_driver extends uvm_driver;
  `uvm_component_utils(ALSU_driver)

  virtual ALSU_if ALSU_vif;

  function new(string name = "ALSU_driver", uvm_component parent = null);
     super.new(name, parent);
  endfunction

  function void build_phase(uvm_phase phase);
    super.build_phase(phase);

    if (!uvm_config_db #(virtual ALSU_if)::get(this , "", "ALSU_IF_comps", ALSU_vif))begin
      `uvm_fatal("build_phase" , "Driver - Unable to get the virtual interface")
    end
  endfunction

  task run_phase(uvm_phase phase);
    super.run_phase(phase);
    ALSU_vif.cin = 0;
    ALSU_vif.red_op_A = 0;
    ALSU_vif.red_op_B = 0;
    ALSU_vif.bypass_A = 0;
    ALSU_vif.bypass_B = 0;
    ALSU_vif.direction = 0;
    ALSU_vif.serial_in = 0;
    ALSU_vif.opcode = 0;
    ALSU_vif.A = 0;
    ALSU_vif.B = 0;
    ALSU_vif.rst = 1;
    @(negedge ALSU_vif.clk);
    ALSU_vif.rst = 0;
    forever begin
      @(negedge ALSU_vif.clk);
      ALSU_vif.cin = $random();
      ALSU_vif.red_op_A = $random();
      ALSU_vif.red_op_B = $random();
      ALSU_vif.bypass_A = $random();
      ALSU_vif.bypass_B = $random();
      ALSU_vif.direction = $random();
      ALSU_vif.serial_in = $random();
      ALSU_vif.opcode = $random();
      ALSU_vif.A = $random();
      ALSU_vif.B = $random();
      ALSU_vif.rst = $random();
    end
  endtask
endclass
```

Kareem Waseem

**Part #3**

```
class ALSU_config extends uvm_object;
  `uvm_object_utils(ALSU_config)

  virtual ALSU_if ALSU_vif;

  function new(string name = "ALSU_config");
    super.new(name);
  endfunction
endclass
```

```
module top();
    bit clk;

    initial begin
        forever
            #1 clk = ~clk;
    end

    ALSU_if ALSUif (clk);
    ALSU DUT (ALSUif.A, ALSUif.B, ALSUif.cin, ALSUif.serial_in, ALSUif.red_op_A, ALSUif.red_

    initial begin
        uvm_config_db#(virtual ALSU_if)::set(null, "uvm_test_top", "ALSU_IF", ALSUif);
        run_test("ALSU_test");
    end

endmodule
```

```
class ALSU_env extends uvm_env;
    `uvm_component_utils(ALSU_env)

  ALSU_driver driver;

  function new(string name = "ALSU_env", uvm_component parent = null);
    super.new(name,parent);
  endfunction

  function void build_phase(uvm_phase phase);
    super.build_phase(phase);
    driver = ALSU_driver::type_id::create("driver",this);
  endfunction: build_phase
endclass
```

Karim Wasem

```
class ALSU_driver extends uvm_driver;
  `uvm_component_utils(ALSU_driver)

  virtual ALSU_if ALSU_vif;
  ALSU_config ALSU_cfg;

  function new(string name = "ALSU_driver", uvm_component parent = null);
     super.new(name, parent);
  endfunction

  function void build_phase(uvm_phase phase);
    super.build_phase(phase);

    if (!uvm_config_db #(ALSU_config)::get(this , "", "CFG", ALSU_cfg))begin
      `uvm_fatal("build_phase" , "Driver - Unable to get configuration object")
    end
  endfunction

  function void connect_phase(uvm_phase phase);
   super.connect_phase(phase);
   ALSU_vif = ALSU_cfg.ALSU_vif;
  endfunction

  task run_phase(uvm_phase phase);
    super.run_phase(phase);
    ALSU_vif.cin = 0;
    ALSU_vif.red_op_A = 0;
    ALSU_vif.red_op_B = 0;
    ALSU_vif.bypass_A = 0;
    ALSU_vif.bypass_B = 0;
    ALSU_vif.direction = 0;
    ALSU_vif.serial_in = 0;
    ALSU_vif.opcode = 0;
    ALSU_vif.A = 0;
    ALSU_vif.B = 0;
    ALSU_vif.rst = 1;
    @(negedge ALSU_vif.clk);
    ALSU_vif.rst = 0;
    forever begin
      @(negedge ALSU_vif.clk);
      ALSU_vif.cin = $random();
      ALSU_vif.red_op_A = $random();
      ALSU_vif.red_op_B = $random();
      ALSU_vif.bypass_A = $random();
      ALSU_vif.bypass_B = $random();
      ALSU_vif.direction = $random();
      ALSU_vif.serial_in = $random();
      ALSU_vif.opcode = $random();
      ALSU_vif.A = $random();
      ALSU_vif.B = $random();
      ALSU_vif.rst = $random();
    end
  endtask
endclass
```

```
class ALSU_test extends uvm_test;
  `uvm_component_utils(ALSU_test)

  ALSU_env env;
  ALSU_config ALSU_cfg;

  function new(string name = "ALSU_test", uvm_component parent = null);
    super.new(name, parent);
  endfunction

  function void build_phase(uvm_phase phase);
    super.build_phase(phase);
    env = ALSU_env::type_id::create("env",this);
    ALSU_cfg = ALSU_config::type_id::create("ALSU_cfg");

    if (!uvm_config_db #(virtual ALSU_if)::get(this, "","ALSU_IF", ALSU_cfg.ALSU_vif))
      `uvm_fatal("build_phase", "Test - Unable to get the virtual interface of the ALSU from the uvm_config_db")

    uvm_config_db #(ALSU_config)::set(this, "*", "CFG", ALSU_cfg);
  endfunction


  task run_phase(uvm_phase phase);
    super.run_phase(phase);
    phase.raise_objection(this);
    #100;`uvm_info("run_phase", "Welcome to the UVM Env.", UVM_MEDIUM)
    phase.drop_objection(this);
  endtask: run_phase
endclass: ALSU_test
```

Kareem Waseem